1) What inspired or motivated you to design and develop the PPH cup? When did you start working on it? When did you build your first functional prototype?

Since my childhood, I had to witness and experience the emotional trauma of maternal deaths; during childbirth, due to excessive blood loss.

I learned from my doctor parents that there is no single ideal method to measure the post delivery blood loss accurately. It leads to delay in diagnosis and unfortunate death of mothers in spite of marathon efforts by the medical team.

This inspired me to find the solution to the problem.

With the knowledge of the science and the menstrual cup, I learned during my science research project on sanitary waste I developed the concept of postpartum cup.

- It was tested for safety and toxicity at Indian Drug research Laboratory, Pune.
- Later the saferty certificate is obtained from USAFDA as well.

The pilot study was conducted to know the acceptance and its safety and efficacy in clinical practice under supervision of obstetricians at 5 private hospitals.

The project was presented to many experts digitally and the feedback was encouraging.

Further study conducted at government medical college, Akola.

Constant feedback is taken from the users for further improvements.

➢ The timeline is as follows:-

- Idea on paper(2018)-
- prototype(2019)-
- design(2020)-
- mould(2020)-
- permission/ethical committee (march2020) –

- rectification n revision(2020)

2) **Could you explain to us in some detail how the PPH cup works? What are some of the standout features of this innovation? What is it made of?**

- The PPHcup is a **silicone** cup.
- Inserted into birth canal before or after caesarean section or after normal delivery. It has 2 ports at the end. One port is for attaching the collection bag. Another port provided facility for suction to avoid blockage of the system.
- It has a potential to be the method of choice for quantification of obstetric blood loss globally. There by reducing the maternal mortality worldwide
- It is **reusable** so avoid generation of biomedical waste.
- The negative pressure system may serve the therapeutic purpose by sucking out the blood clots in cases of atomic pph .
- Economically viable and environmentally sustainable.
- Cash free, rash free and trash free.

3) **What role did your parents and peers play in coming up with this innovation?**

- My parents encouraged me to peruse the idea; they provided all the support including financial support needed to take the innovation from the idea through incubation to actual product and networking with the professionals. I got very limited help from the peers.

- My school principle Mr.Ganesh Patil and science project mentor Mrs.Sangeeta Khanna from Kashinath Palod Public school ,Jalgaon also guided me and provided opportunities to participate in various national and international science fair like INSPIRE,IRIS,ISEF.

4) **Could you take us through in some detail the process of how you designed and developed the PPH Cup? What were some of the key trials and errors you overcame before finally landing upon a successful prototype?**

I used the spare menstrual cup of my mother to prepare the prototype model. It was presented to the obstetricians .They liked the idea and encourage me further. The PPH cup was designed and mould was developed.

The pilot study was conducted to know the acceptance and its safety and efficacy in clinical practice under supervision of obstetricians at 5 private hospitals.

The project was presented to many experts digitally and the feedback was encouraging.

Further study is going on at Government medical college, Akola. Government medical college Aurangabad, Mahatma Gandhi Institute of Medical Sciences, Sewagram (Wardha) in Maharashtra.

Now nearly 50 hospitals across India are the users.

Constant feedback is taken from the users for further improvements.

- The mould was revised for rectification and now second version of the mould is developed.
- The prototype and mould development was partly funded by the Inspire, DST,Government of India

5) **Could you describe to us the testing process the PPH Cup went through? Are practicing doctors currently deploying it in their regular practice?**

The safety was tested by

- It was tested for safety and toxicity at Indian Drug research Laboratory, Pune.
- Later the safety certificate is obtained from USAFDA as well.
- Further study is going on at Government medical college, Akola. Government medical college Aurangabad, Mahatma Gandhi Institute of Medical Sciences, Sewagram
- Now nearly 50 hospitals across India are the users.
- It has been used in more than thousands of patients and saved many precious life of the mother undergoing child birth due to prompt diagnosis and early medical treatment.

6) **How much does it cost? Have people in the industry contacted you for mass production? If not, do you have any plans of mass producing it in the future?**

- The net production cost is Rs 500 (excluding developing cost and human hours invested for it)
- I was not contacted by any company from the industry.
- I have formed Archit scientific research private limited company.
- I have plans to collaborate with government of India, biosciences department of IIT and the medical industry to get super finished product and mass production.
- The innovation has been applied for the patent and published but yet to get the final patent.
- I hold the IP rights like trademark, copyright of logo, and name of the product

7) **How old are you currently? Where are you studying and in what class? What are your plans for the future?**

- I am 15 year 6 months old.

- I am studying at Kashinath Palod Public school, Jalgaon, Maharashtra. I am sophomore (class 10).
- I want to learn about AI and machine learning as well as python. After my high school graduation, I want to get into the best universities of the world to explore my academic and professional interest for my future career
- My aim is to represent team India in science and mathematics Olympiads. I dream to study in best university and serve humanity. I wish to work to make the world, better place to live in.

8) Do you have a message for other child innovators around the country?

Failures and criticism are a part of every journey and suggests that one should not stop for the fear of criticism. “Take it positively, work hard, learn from mistakes, pursue your goal and never ever stop, don’t let the success go to your head.

We need to think of using technology to come up with smart, sustainable solutions but at the same time try and avoid waste."

Please see video at the bottom in gallery section in the following link.

https://www.architpphcup.com/gallery